

Dimensions of Tetrahedral Z -Cevian Families

Face constraints, jaws, tripods, and odd-dimensional obstructions

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Abstract

Let $g(a, b, c)$ be a nonzero symmetric, homogeneous, real-analytic function of the side lengths of a triangle, with $g(1, 1, 1) = 0$, and let $\mathcal{S}_g$ be the locus of nondegenerate tetrahedra whose four faces all satisfy $g = 0$. We study the dimension of such four-face lifts and their relation to tetrahedral Z -cevia concurrence.

Two general dimension theorems are proved. First, $\dim \mathcal{S}_g \leq 3$, with scale included; thus a single nontrivial analytic face constraint cannot produce a four- or five-dimensional tetrahedral family. Second, a proper real-analytic Z -cevia family whose facial barycentric weights are nonvanishing has no five-dimensional analytic component. The second obstruction comes from an omitted-opposite-edge property of the six compatibility residuals. In the opposite direction we give an explicit real-analytic triangle center whose Z -cevia locus has a genuine three-dimensional component containing the jaws, with an exact Jacobian certificate. Together with the universal two-dimensional kite locus known from earlier work, these results substantially sharpen the odd-dimensionality question posed for tetrahedral Z -cevians.

For the isosceles constraint

$$g_1 = \frac{(a-b)^2(b-c)^2(c-a)^2}{(a+b+c)^6},$$

we give a complete edge-length classification. Its maximal three-dimensional components are exactly the three labeled jaw families and the twelve labeled tripod families; six two-dimensional alternating-path components are additionally required for the full set-theoretic decomposition. Four further natural constraints give two-dimensional facewise families. A sixth constraint, detecting equilateral triangles or a 2:1 side ratio, has dimension three and contains a natural family of 2-gearred jaws.

Finally, for a broad class of centers that reduce to the centroid on $g = 0$, we prove the guaranteed inclusion $\mathcal{S}_g \subseteq \text{ZCevian}[Z]$, isolate the accidental-concurrence loophole, and state a conditional generic transversality result. The finite branch calculations are reproduced by one short exact Sage certificate for the all-isosceles reduction and the $g_2, \dots, g_5$ positive-chart branch dimensions.

Keywords. Triangle center; tetrahedron; cevian concurrence; face constraint; analytic set; dimension; jaw; tripod; Gröbner basis; Encyclopedia of Triangle Centers.

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1 Introduction

A triangle center is usually studied inside one triangle. The present paper uses it instead as a rule imposed simultaneously on the four faces of a tetrahedron. Given a triangle center Z , one joins each tetrahedral vertex to the copy of Z in the opposite face. The resulting four lines are the Z -cevians. Their concurrence is a genuinely spatial compatibility condition: the three barycentric weights visible in each face must be restrictions of one set of four global weights.

This problem continues the program initiated in Hajja–Hammoudeh–Hayajneh–Martini [17]. That paper asked, in particular, what odd dimensions can occur for tetrahedral Z -cevia families.

It also showed that every such family contains the two-dimensional kite locus, so dimension one is unavailable in the usual global sense. The present paper settles the five-dimensional part for real-analytic center rules with nonvanishing facial weights and exhibits genuine three-dimensional analytic components. We deliberately distinguish this local-component statement from the stronger assertion that a particular explicit center has no remote four-dimensional component; the latter is a global elimination problem.

A companion article introduces the four-dimensional power- k families and their parallel Cevian-pedal theory [18]. Two further companion papers study facial projection rather than concurrence: the tetrahedral Euler object [19] and the skeletal-centroid line [20]. Thus the present article occupies the dimension-theoretic part of a broader study of how planar triangle centers constrain tetrahedral geometry.

This viewpoint connects three subjects which are usually treated separately.

- (i) A symmetric equation $g(a, b, c) = 0$ selects a similarity class, or a curve of similarity classes, of triangles.
- (ii) Requiring all four faces of a tetrahedron to satisfy the same equation defines a subset $\mathcal{S}_g$ of the six-dimensional edge-length space.
- (iii) If a triangle center becomes the centroid whenever $g = 0$, then every tetrahedron in $\mathcal{S}_g$ automatically has concurrent Z -cevians.

The last assertion follows because the four ordinary medians of every tetrahedron concur at its centroid.

The simple construction

$$F_a(a, b, c) = 1 + \lambda \frac{a}{a + b + c} g(a, b, c)$$

therefore converts a facial constraint into infinitely many triangle centers. It is tempting to identify the resulting Z -cevian family with $\mathcal{S}_g$. That identification is generally too strong. Cevian concurrence compares ratios of weights on adjacent faces; those ratios can agree even when the four values of g are nonzero. We call this phenomenon *accidental concurrence*. A major purpose of this paper is to keep the guaranteed facewise locus separate from any additional concurrence components.

The dimension question is especially informative. A tetrahedron has six edge lengths. Four generic scalar face equations suggest dimension $6 - 4 = 2$, including scale. Dependencies among the four conditions can raise the dimension. The all-isosceles constraint does so: its largest components have dimension three. We prove that three is the absolute maximum for a nontrivial symmetric homogeneous analytic face constraint. Thus dimensions four and five cannot be obtained by a single equation imposed independently on all four faces.

The all-isosceles case also exposes a useful terminological distinction. The three-dimensional part consists exactly of jaws and tripods, but the complete set has six additional two-dimensional components. Accordingly, the mathematically correct “Jaws-Tripods theorem” is a theorem about maximal-dimensional components, followed by a short list of lower-dimensional residual strata.

Triangle centers are understood in the functional sense introduced by Kimberling [14, 15]; the online Encyclopedia of Triangle Centers (ETC) provides the standard catalogue [5, 16]. Our algebraic computations use standard Gröbner-basis and dimension methods [3, 6]. Since positivity, triangle inequalities, and the Cayley-Menger inequality are semialgebraic open conditions, they select physically realizable parts of algebraic components without increasing their dimension; see [2].

2 Definitions and the basic inclusion

2.1 The edge cone

Label the edges of a tetrahedron $T = V_1V_2V_3V_4$ by

$$x_{ij} = |V_iV_j|, \quad 1 \leq i < j \leq 4.$$

Let $\mathcal{E} \subset \mathbb{R}_{>0}^6$ be the set of edge vectors for which every facial triple satisfies the strict triangle inequalities and the Cayley–Menger determinant is positive. Thus $\mathcal{E}$ is the open semialgebraic cone of nondegenerate Euclidean tetrahedra, modulo rigid motion but not modulo scale. Every dimension in this paper includes scale. Passing to similarity classes subtracts one.

The four faces, written only in terms of their edges, are

$$\begin{aligned} F_1 &= (x_{23}, x_{24}, x_{34}), & F_2 &= (x_{13}, x_{14}, x_{34}), \\ F_3 &= (x_{12}, x_{14}, x_{24}), & F_4 &= (x_{12}, x_{13}, x_{23}). \end{aligned} \tag{1}$$

Definition 2.1 (Analytic facial constraint). An *analytic facial constraint* is a nonzero real-analytic function g on the positive triangle cone such that

$$g(a, b, c) = g(a, c, b) = g(b, a, c), \quad g(ta, tb, tc) = g(a, b, c)$$

for $t > 0$, and $g(1, 1, 1) = 0$. Its face-constrained tetrahedral locus is

$$\mathcal{S}_g = \{x \in \mathcal{E} : g(F_1) = g(F_2) = g(F_3) = g(F_4) = 0\}.$$

The symmetry requirement makes the equation independent of the labeling of a face. Degree zero means that it is a condition on the shape of a triangle. A rational homogeneous expression is analytic on the positive cone whenever its denominator has no zero there.

2.2 Triangle-center functions and Z -cevians

Let $F(a, b, c)$ be a homogeneous, bisymmetric barycentric center function in Kimberling’s sense:

$$F(a, b, c) = F(a, c, b), \quad F(ta, tb, tc) = t^d F(a, b, c).$$

The corresponding center of a triangle with side lengths a, b, c has barycentrics

$$Z_F = (F(a, b, c) : F(b, c, a) : F(c, a, b)).$$

Definition 2.2 (Z -cevia family). For each i , place Z_F in the face F_i opposite V_i . The line joining V_i to this facial point is the i -th Z -cevia. The family $\text{ZCevian}[Z_F]$ consists of the tetrahedra in $\mathcal{E}$ for which all four Z -cevians concur.

For a constraint g , a positive constant C , and a real parameter λ , define

$$F_a^{g, \lambda}(a, b, c) = C + \lambda \frac{a}{a + b + c} g(a, b, c). \tag{2}$$

The other two weights are obtained cyclically. Because g is symmetric and homogeneous of degree zero, this is a homogeneous bisymmetric center function. If positivity of all weights is desired, one may restrict λ or the normalized shape domain. Projective barycentrics themselves do not require positivity.

Proposition 2.3 (Guaranteed median component). *For every analytic facial constraint g and every λ ,*

$$\mathcal{S}_g \subseteq \text{ZCevian}[X_{g, \lambda}],$$

where $X_{g, \lambda}$ is defined by (2).

Proof. If a face satisfies $g = 0$, its three weights in (2) are all equal to C . The facial center is therefore its centroid. If all four faces satisfy $g = 0$, the four $X_{g,\lambda}$ -cevians are the four tetrahedral medians, which concur at

$$G_T = \frac{V_1 + V_2 + V_3 + V_4}{4}.$$

□

Remark 2.4 (The correct dimensional wording). The set of triangles satisfying $g = 0$ is not the object whose dimension should be compared directly with a tetrahedral Z -cevia family. The appropriate object is its *four-face lift* $\mathcal{S}_g$. Proposition 2.3 gives

$$\dim \text{ZCevian}[X_{g,\lambda}] \geq \dim \mathcal{S}_g.$$

Equality requires an additional assertion excluding maximal-dimensional accidental-concurrence components.

3 Two general dimension theorems

3.1 A sharp upper bound for $\mathcal{S}_g$

We first isolate the elementary linear-algebra fact behind the bound. The six coordinates of $\mathbb{R}^6$ are indexed by the edges of K_4 ; the four facial triples are its four triangles.

Lemma 3.1 (Four triangular supports). *For $i = 1, \dots, 4$, let $r_i \in \mathbb{R}^6$ be nonzero and supported on the three edges of the face F_i . If every r_i has at least two nonzero coordinates, then*

$$\text{rank}(r_1, r_2, r_3, r_4) \geq 3.$$

Proof. Any two facial supports meet in exactly one edge. Since each vector has at least two nonzero coordinates, no two vectors belonging to different faces can be proportional. Suppose, for contradiction, that all four vectors span a plane. Use r_1, r_2 as a basis.

The support of r_1 is $\{23, 24, 34\}$, and that of r_2 is $\{13, 14, 34\}$. To form a vector supported on $F_3 = \{12, 14, 24\}$, a linear combination of r_1, r_2 must eliminate the private coordinates 23 and 13, as well as the shared coordinate 34. A nonzero combination can do this only in the exceptional support pattern

$$\text{supp } r_1 = \{24, 34\}, \quad \text{supp } r_2 = \{14, 34\},$$

up to interchanging the two faces and their private edges. In that case the resulting vector is supported on $\{14, 24\}$. But the entire span is then supported on $\{14, 24, 34\}$, which is disjoint from $F_4 = \{12, 13, 23\}$. It cannot contain the nonzero vector r_4 . The remaining support cases force one coefficient to vanish and are even more immediate. Hence the rank is at least three. □

Theorem 3.2 (No four- or five-dimensional single-constraint lift). *Let g be a nonzero symmetric homogeneous real-analytic facial constraint. Then*

$$\dim \mathcal{S}_g \leq 3.$$

Consequently, no such single face constraint has a four- or five-dimensional four-face lift.

Proof. Normalize a facial triple by its perimeter. Let

$$\Delta = \{(a, b, c) \in \mathbb{R}_{>0}^3 : a + b + c = 1\}, \quad A = \{g = 0\} \cap \Delta.$$

The set A is real analytic in the two-dimensional open simplex Δ , and therefore admits a locally finite smooth stratification [21, 13, 2]. Cone each stratum explicitly:

$$\hat{S} = \{rx : r > 0, x \in S\}.$$

Because g is homogeneous, these coned strata form a smooth scale-invariant stratification of the positive facial zero set. Thus scale invariance is built into the stratification rather than assumed for an arbitrary Whitney stratification.

At a positive point (a, b, c) of a coned stratum, the radial vector (a, b, c) belongs to its tangent space. Every nonzero conormal $\nu = (\nu_a, \nu_b, \nu_c)$ therefore satisfies

$$a\nu_a + b\nu_b + c\nu_c = 0.$$

Since $a, b, c > 0$, such a conormal cannot have only one nonzero coordinate; it has support on at least two sides. A higher-codimension facial stratum, such as the equilateral ray, supplies at least two independent conormals and can only strengthen the estimate.

Pull the four facial stratifications back to $\mathcal{E}$, and take a common smooth refinement of their intersections; equivalently, refine $\mathcal{S}_g$ according to the coned stratum occupied by each of its four faces. Pull the corresponding facial conormals back to the six edge coordinates using (1). If all four selected facial strata have codimension one, Lemma 3.1 shows that their combined rank is at least three. If one face lies on a higher-codimension stratum, choose two independent conormals from that face and one nonzero conormal from a different face. The first two are supported on the same three-edge face and are independent. The support of the third face meets that three-edge set in only one edge, while every nonzero conormal has at least two nonzero coordinates; hence the third conormal has a coordinate outside the first face and is independent of the first two. Thus the combined rank is again at least three.

The tangent space of every smooth stratum of the resulting four-face intersection is contained in the common kernel of these pulled-back conormals. Every such stratum therefore has codimension at least three in $\mathcal{E}$, and hence dimension at most three. $\square$

Remark 3.3. The hypotheses matter. The identically zero function imposes no condition and gives dimension six. A nonhomogeneous equation can fix a side length and destroy the conormal argument based on Euler's identity. Several independent face constraints are a different problem.

3.2 The dimension-five obstruction on the nonvanishing-weight domain

Write w_i^{ijk} for the barycentric weight assigned to V_i by the center Z in the face $V_i V_j V_k$. The four cevians concur if and only if there exists a global weight vector, not identically zero, (W_1, W_2, W_3, W_4) whose restrictions to every face are proportional to its three local weights. In particular, for the two faces ijk and ijl containing the edge ij , one must have

$$R_{ij} := w_i^{ijk} w_j^{ijl} - w_j^{ijk} w_i^{ijl} = 0. \quad (3)$$

The crucial observation is

$$R_{ij} \text{ is independent of } x_{kl},$$

the edge opposite ij . The six equations $R_{ij} = 0$ are redundant, but their common zero set is exactly the compatibility locus wherever the weights are defined and nonzero.

Lemma 3.4 (Cylinder lemma). *Let $U = U_0 \times I \subset \mathbb{R}^{n-1} \times \mathbb{R}$, and let $F(x_1, \dots, x_{n-1})$ be a nonzero real-analytic function that is independent of x_n . Every irreducible codimension-one analytic component of $Z(F) \cap U$ is locally a product $H_0 \times I$. Consequently, if an irreducible analytic hypersurface $H \subset U$ is contained in $Z(F)$, then near a generic point of H it is cylindrical in the x_n -direction.*

Proof. One has $Z(F) \cap U = (Z(F) \cap U_0) \times I$. The maximal-dimensional irreducible components are therefore products of irreducible components in U_0 with I . If an irreducible hypersurface H is contained in $Z(F)$, then it has the same local dimension as one of these maximal components and hence agrees with it on a nonempty relatively open subset. This gives the asserted local product structure at generic points. $\square$

Theorem 3.5 (The dimension-five obstruction). *Let Z have a real-analytic barycentric center function $f(a, b, c)$ that is nowhere zero on the positive triangle cone. If $\text{ZCevian}[Z]$ is a proper subset of $\mathcal{E}$, then it has no five-dimensional analytic component. In particular,*

$$\dim \text{ZCevian}[Z] \neq 5.$$

The same conclusion holds on any connected relabeling-invariant open subdomain on which all cyclic facial weights are real analytic and nonzero.

Proof. The six residuals are carried into one another by relabeling the vertices. If one residual, say R_{12} , were identically zero, then all six would be identically zero. Because all facial weights are nonzero, the six vanishing pairwise cross-products make the local weight ratios on adjacent faces universally compatible. They therefore glue to a global weight vector for every tetrahedron, so the four Z -cevians would concur identically. This would give $\text{ZCevian}[Z] = \mathcal{E}$, contrary to properness. Hence none of the six residuals is identically zero.

Assume that H is an irreducible five-dimensional analytic component of $\text{ZCevian}[Z]$. Then H is locally a hypersurface in the six edge variables and is contained in every zero set $R_{ij} = 0$. Since R_{12} is independent of x_{34} , Lemma 3.4 shows that, near a generic point of H , the hypersurface is cylindrical in the x_{34} -direction. The residual R_{34} omits x_{12} , and the four remaining residuals similarly supply the other four coordinate directions. Because there are only six exceptional analytic subsets, one may choose a single smooth point of H that is generic for all six applications. At that point every coordinate basis vector belongs to TH , forcing $\dim TH \geq 6$, contrary to $\dim H = 5$. $\square$

Remark 3.6. The nonvanishing hypothesis is used only to turn identically vanishing edge residuals into universal compatibility. Center rules with persistent zero barycentric weights may have additional lower-dimensional singular strata and require a separate gluing argument; no claim about such zero-weight strata is made here.

Remark 3.7. The theorem excludes dimension five, not dimension four. Power- k tetrahedra and the isogonic Eisenstein family provide important four-dimensional Z -cevia families [18]. Nor does the theorem force even dimension: the next result gives an explicit analytic three-dimensional component.

3.3 An explicit three-dimensional analytic component

Define the homogeneous bisymmetric center function

$$f_*(a, b, c) = (a + b + c)^4 + \frac{1}{100}(a - b)(a - c)(b - c)^2. \quad (4)$$

It is symmetric in b, c , and therefore defines a real-analytic triangle center in Kimberling's functional sense. It is also strictly positive on the positive triangle cone: each of the four difference factors has absolute value less than $a + b + c$, so the perturbation has absolute value less than $(a + b + c)^4/100$.

Theorem 3.8 (A genuine three-dimensional Z -cevia component). *The Z -cevia locus associated with (4) has a genuine three-dimensional component containing the four-cycle jaw family.*

Proof. On an isosceles triangle, each of the three cyclic perturbation terms in (4) vanishes. Thus the corresponding center is the centroid of every isosceles triangle. Every face of a jaw is isosceles, so all four Z_{f_*} -cevians are medians and concur at the tetrahedral centroid. Hence the three-parameter jaw manifold

$$x_{12} = x_{23} = x_{34} = x_{14} = L, \quad x_{13} = p, \quad x_{24} = q$$

is contained in the concurrence locus.

To certify the local dimension, form the six compatibility residuals (3) and differentiate with respect to $(x_{12}, x_{13}, x_{14}, x_{23}, x_{24}, x_{34})$. At the admissible jaw $(L, p, q) = (5, 6, 7)$, the Cayley–Menger determinant gives

$$V^2 = \frac{735}{4} > 0,$$

and the residual vector vanishes exactly. The submatrix formed by residuals R_{12}, R_{13}, R_{14} and columns x_{12}, x_{14}, x_{23} is

$$J_0 = \begin{pmatrix} 607809/100 & -131072/25 & -83521/100 \\ 32768/25 & -32768/25 & -32768/25 \\ 131072/25 & -607809/100 & 0 \end{pmatrix}, \quad \det J_0 = -\frac{179359981764608}{15625} \neq 0.$$

Thus the full residual Jacobian has rank at least three. On the other hand, the residual map vanishes identically on the smooth three-parameter jaw manifold, so its differential annihilates the three-dimensional jaw tangent space and has rank at most three. Consequently its rank is exactly three. Three independent residuals cut out a local manifold of dimension $6 - 3 = 3$ containing the jaw manifold; the full concurrence locus is contained in that manifold and already contains the jaw manifold. Hence the local component through this point is exactly three-dimensional. $\square$

Remark 3.9 (What this does and does not settle). Theorem 3.8 proves that odd-dimensional analytic components really occur, so the dimensions seen in ETC computations need not be even. It does not by itself exclude a separate four-dimensional component of the same center’s global concurrence locus. Establishing global dimension exactly three for a particular simple explicit center requires a saturated elimination or an equivalent global certificate. This distinction is important in interpreting the odd-dimensionality question from [17].

4 The isosceles constraint and the Jaws–Tripods theorem

Let $s = a + b + c$, and set

$$g_1(a, b, c) = \frac{(a - b)^2(b - c)^2(c - a)^2}{s^6}. \quad (5)$$

Then $g_1 = 0$ if and only if the triangle is isosceles, with equilateral triangles included.

Definition 4.1 (Jaw). A *jaw* is a tetrahedron whose vertices can be cyclically labeled so that four cycle edges are equal:

$$x_{12} = x_{23} = x_{34} = x_{14} = L,$$

while the two diagonals $x_{13} = p$ and $x_{24} = q$ are free, subject to the Euclidean inequalities.

Definition 4.2 (Tripod). A *tripod* is a tetrahedron with three equal edges incident at one vertex and with an isosceles opposite face. For example,

$$x_{12} = x_{13} = x_{14} = L, \quad x_{23} = x_{24} = p, \quad x_{34} = q.$$

There are four choices of the distinguished vertex and three choices of the equal pair in the opposite face.

Definition 4.3 (Alternating-path stratum). An *alternating-path* tetrahedron has, after relabeling,

$$x_{12} = x_{13} = x_{24} = u, \quad x_{14} = x_{23} = x_{34} = v. \quad (6)$$

Each of the two length classes forms a three-edge path on the four vertices. There are six labeled components.

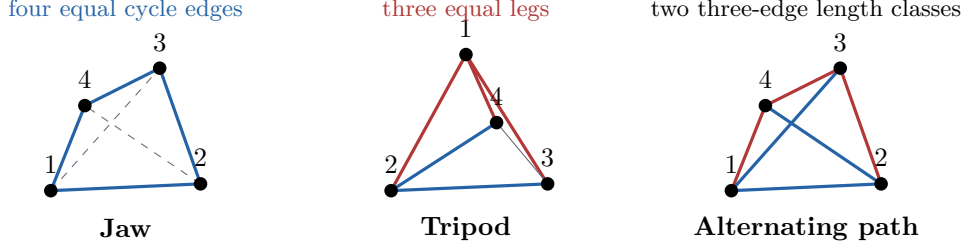


Figure 1: The maximal three-dimensional all-isosceles components are jaws and tripods. Six two-dimensional alternating-path components are also needed for the full set-theoretic classification.

Theorem 4.4 (Jaws–Tripods theorem, complete form). *The all-isosceles tetrahedral locus $\mathcal{S}_{g_1}$ has dimension three. Its maximal three-dimensional linear components in edge-length space are exactly*

- (i) *three labeled jaw components;*
- (ii) *twelve labeled tripod components.*

The irredundant set-theoretic decomposition additionally contains six two-dimensional alternating-path components of the form (6). Thus the phrase “exactly the union of jaws and tripods” is correct for the maximal-dimensional part, but not for the full locus unless the six residual strata are mentioned.

Proof. Every face must choose one of its three side equalities. Hence $\mathcal{S}_{g_1}$ is the union of $3^4 = 81$ linear branch assignments. A branch is represented by a 4×6 matrix with one row $x_e - x_f = 0$ for each face.

We first classify the rank-three branches without computation. At a fixed vertex, inspect its three incident edges.

If all three are equal, the opposite face must be isosceles; this is a tripod. If exactly two are equal, say $x_{12} = x_{13} \neq x_{14}$, the two faces containing x_{14} force

$$x_{24} \in \{x_{12}, x_{14}\}, \quad x_{34} \in \{x_{13}, x_{14}\}.$$

If both choices use the first length, four edges form an equal cycle, giving a jaw. If both use the second, one obtains a tripod centered at V_4 . In the mixed cases the fourth face either forces a tripod at V_2 or V_3 , or introduces a fourth independent equality, leaving a two-dimensional branch.

If the three incident edges are pairwise distinct, each opposite edge is forced to equal one of the two adjacent incident edges. The fourth face can be isosceles only when two of these choices agree. This produces a tripod centered at one of the other vertices, unless four independent equalities are created. Therefore every rank-three branch is a jaw or a tripod.

Conversely, each jaw and each tripod makes all four faces isosceles and is defined by three independent linear equalities. There are three Hamiltonian cycles in K_4 , hence three labeled jaws. There are $4 \cdot 3 = 12$ labeled tripods.

For completeness, exact row reduction of all 81 assignments gives

matrix rank	3	4
branch assignments	15	66
distinct linear spaces	15	24.

Eighteen of the distinct rank-four spaces lie inside one of the rank-three components. The remaining six are precisely the relabelings of (6). This proves the irredundant decomposition.

Finally, the listed components meet the nondegenerate Euclidean cone in relatively open sets. For example, a tripod with $(L, p, q) = (3, 4, 5)$ is realizable, with exact value

$$V^2 = \frac{2375}{144} > 0.$$

For an alternating-path component, $(u, v) = (1, 3/2)$ has

$$V^2 = \frac{143}{9216} > 0.$$

Hence the largest realized components really have dimension three. $\square$

5 Four generic two-dimensional constraints

Let $s = a + b + c$, and define cyclically

$$Q_A = b^2 + c^2 - a^2 - bc, \quad (7)$$

$$M_A = 2a(b^2 + c^2 - a^2) - b(c^2 + a^2 - b^2) - c(a^2 + b^2 - c^2). \quad (8)$$

The following symmetric degree-zero functions encode four natural disjunctive constraints:

$$g_2 = \frac{Q_A Q_B Q_C}{s^6}, \quad (9)$$

$$g_3 = \frac{M_A M_B M_C}{s^9}, \quad (10)$$

$$g_4 = \frac{(2a - b - c)(2b - c - a)(2c - a - b)}{s^3}, \quad (11)$$

$$g_5 = \frac{(2a^2 - b^2 - c^2)(2b^2 - c^2 - a^2)(2c^2 - a^2 - b^2)}{s^6}. \quad (12)$$

Table 1: The four constraints and their exact positive-chart dimensions. The chart dimension is computed after setting $x_{34} = 1$. Because every point of $\mathcal{E}$ has $x_{34} > 0$, adjoining scale gives the positive-cone dimension.

	Geometric meaning of one branch	branches	chart dim.	cone dim.
g_2	one angle is 60°	$3^4 = 81$	1	2
g_3	one angle cosine is the mean of the other two	81	1	2
g_4	one side is the arithmetic mean of the other two	81	1	2
g_5	one squared side is the mean of the other two squares	81	1	2

Indeed, $Q_A = 0$ is the cosine-law equation $\cos A = 1/2$. Multiplying $2 \cos A - \cos B - \cos C = 0$ by $2abc$ gives $M_A = 0$. The meanings of g_4 and g_5 are immediate.

Proposition 5.1. *For each $i = 2, 3, 4, 5$, every one of the 81 branch systems has Krull dimension one in the chart $x_{34} = 1$. Consequently, every branch meeting $\mathcal{E}$ has dimension at most two after scale is restored. Moreover, $\mathcal{S}_{g_i} \cap \mathcal{E}$ contains a two-dimensional Euclidean subfamily. Hence*

$$\dim \mathcal{S}_{g_i} = 2, \quad i = 2, 3, 4, 5.$$

Exact computational proof. Each face has three possible cyclic branches, so each g_i gives 81 branch systems. We dehomogenized by $x_{34} = 1$ and computed the Krull dimension of every branch ideal over $\mathbb{Q}$, using a degree-reverse-lexicographic Gröbner basis. For each of g_2, g_3, g_4, g_5 , all 81 ideals had dimension one in the chart. Because every Euclidean edge vector has $x_{34} > 0$, restoring scale gives the required upper bound two on the positive Euclidean locus. The summary is Table 1.

This is an exact symbolic calculation, not a numerical rank test. The short reproducing program is given in Appendix A.

For the lower bound, choose a nontrivial smooth one-parameter family of acute triangles on one branch of the corresponding constraint. Such families are immediate for g_2, g_4, g_5 ; for g_3 , in angle coordinates, put

$$F(B, C) = \cos(\pi - B - C) - \frac{\cos B + \cos C}{2}.$$

At the equilateral point,

$$\left. \frac{\partial F}{\partial B} \right|_{B=C=\pi/3} = \frac{3\sqrt{3}}{4} \neq 0,$$

so the implicit-function theorem supplies the required acute family. Every acute triangle is the face of a disphenoid whose four faces are congruent to it: in the standard $(\pm x, \pm y, \pm z)$ realization one may take, cyclically,

$$x^2 = \frac{b^2 + c^2 - a^2}{8} > 0.$$

The shape parameter together with scale therefore gives a two-dimensional Euclidean subfamily of each $\mathcal{S}_{g_i}$. $\square$

Remark 5.2. Some individual branch ideals can have no positive Euclidean point, and distinct branch assignments can describe the same component. Neither issue can increase dimension; the disphenoid construction above supplies the needed global Euclidean lower bound. The calculation makes no claim about algebraic components confined to the coordinate hyperplane $x_{34} = 0$; such components cannot meet $\mathcal{E}$ and are irrelevant to the tetrahedral locus studied here.

6 A sixth constraint: 2:1 gearing

Put

$$E = (a - b)^2 + (b - c)^2 + (c - a)^2$$

and

$$R_2 = (a - 2b)(b - 2a)(b - 2c)(c - 2b)(c - 2a)(a - 2c).$$

Define

$$g_6(a, b, c) = \frac{E R_2^2}{s^{14}}. \quad (13)$$

The function is symmetric, nonnegative, homogeneous of degree zero, and analytic on the positive triangle cone. Moreover,

$$g_6 = 0 \iff \text{the triangle is equilateral, or one side is twice another.}$$

Unlike g_1 , it does not vanish on a generic isosceles triangle.

Definition 6.1 (2-geared jaw). A 2-geared jaw is a tetrahedron with a cyclic labeling such that

$$x_{12} = x_{34} = L, \quad x_{23} = x_{14} = 2L, \quad x_{13} = p, \quad x_{24} = q. \quad (14)$$

Every face of (14) contains the pair $L, 2L$, so every face satisfies $g_6 = 0$.

Theorem 6.2 (Exact dimension of the g_6 -locus). *The face-constrained family $\mathcal{S}_{g_6}$ has dimension three. It contains the 2-gearred jaws (14) as a natural three-dimensional subfamily.*

Proof. The function g_6 is a nonzero symmetric homogeneous real-analytic constraint on the positive triangle cone. Theorem 3.2 therefore gives

$$\dim \mathcal{S}_{g_6} \leq 3.$$

The geared-jaw equations (14) have three independent parameters L, p, q , so they supply the matching lower bound whenever they meet the nondegenerate Euclidean cone. They do: taking

$$L = 1, \quad p = q = 2$$

gives

$$(x_{12}, x_{13}, x_{14}, x_{23}, x_{24}, x_{34}) = (1, 2, 2, 2, 2, 1),$$

whose Cayley–Menger evaluation is

$$V^2 = \frac{7}{72} > 0.$$

Hence $\dim \mathcal{S}_{g_6} = 3$. No enumeration of the 7^4 facial branch assignments is required. $\square$

Remark 6.3 (A ratio parameter). For $\rho > 0$, one may replace 2 by ρ :

$$R_\rho = \prod_{\{x,y\} \subset \{a,b,c\}} (x - \rho y)(y - \rho x), \quad g_\rho = \frac{E R_\rho^2}{s^{14}}.$$

The cyclic pattern $L, \rho L, L, \rho L$, with two free diagonals, again supplies a three-parameter face-wise family whenever it has a nonempty Euclidean part. Theorem 3.2 supplies the matching upper bound for every fixed ρ , so such a locus has dimension exactly three. Special values may merge facial branches, but they cannot raise the dimension.

7 From $\mathcal{S}_g$ to Z -cevia families

7.1 The accidental-concurrence loophole

For the canonical center $X_{g,\lambda}$, the condition $g = 0$ turns every facial center into the centroid. Proposition 2.3 therefore gives a three-dimensional component whenever $\dim \mathcal{S}_g = 3$. It does not give equality.

To see the source of the loophole, let $g_i = g(F_i)$. The edge residual (3) compares rational expressions in the weights on two adjacent faces. It can vanish because $g_i = g_j = 0$, but it can also vanish because two nontrivial ratios happen to agree. Four such agreements may define a component not contained in $\mathcal{S}_g$. Therefore one must distinguish:

$$\boxed{\mathcal{S}_g \subseteq \text{ZCevian}[X_{g,\lambda}]} \quad \text{from} \quad \boxed{\mathcal{S}_g = \text{ZCevian}[X_{g,\lambda}]}.$$

The first statement is automatic; the second requires elimination, primary decomposition, or an equivalent global certificate.

For g_1 and g_6 , we can conclude unconditionally that

$$\dim \text{ZCevian}[X_{g_i,\lambda}] \geq 3 \quad (\lambda \neq 0).$$

On any connected relabeling-invariant regular domain where the facial weights are nonzero, Theorem 3.5 excludes dimension five. A proper family may nevertheless have dimension three or four. The centroid $\lambda = 0$ is universal and has dimension six.

7.2 What an infinite set of centers does and does not prove

There are infinitely many center functions that collapse to the centroid on $g = 0$:

$$F_a = C + h(a, b, c) g(a, b, c), \quad (15)$$

where the cyclically generated multiplier is analytic and the resulting center function is homogeneous and bisymmetric. Positivity of h may be imposed when it is geometrically desirable; it is not needed for Proposition 2.3.

Cardinality alone does *not* prove that one member has Z -cevia dimension exactly three. It is logically possible, in the absence of further information, that every member of a one-parameter family possesses a four-dimensional accidental component. Within the nonvanishing-weight scope of Theorem 3.5, the impossibility of dimension five does not rule this out.

The correct generic statement uses transversality.

Proposition 7.1 (Conditional generic transversality bound). *Let $\mathcal{H}$ be a finite-dimensional real-analytic parameter manifold of equivariant multipliers in (15). For each parameter h , form the four necessary edge residuals*

$$\mathcal{R}_h = (R_{12}^{(h)}, R_{13}^{(h)}, R_{14}^{(h)}, R_{23}^{(h)}),$$

and let

$$\mathcal{R} : (\mathcal{E} \setminus \mathcal{S}_g) \times \mathcal{H} \longrightarrow \mathbb{R}^4, \quad (x, h) \longmapsto \mathcal{R}_h(x).$$

Suppose that $\mathcal{R}$ is transverse to 0. Then for a residual set of $h \in \mathcal{H}$, the accidental-concurrence locus has dimension at most two. Consequently, if $\dim \mathcal{S}_g = 3$, then

$$\dim \text{ZCevian}[X_{g,h}] = 3$$

for generic h , provided the family is evaluated on the regular domain of its barycentric weights.

Proof. Every accidental-concurrence point satisfies all six edge residuals, and hence in particular

$$(\text{ZCevian}[X_{g,h}] \setminus \mathcal{S}_g) \subseteq \mathcal{R}_h^{-1}(0).$$

No sufficiency claim for the chosen four residuals is needed. By the parametric transversality theorem [7, 12], a generic partial map $\mathcal{R}_h : \mathcal{E} \setminus \mathcal{S}_g \rightarrow \mathbb{R}^4$ is transverse to 0. Its zero set is therefore either empty or a smooth set of dimension $6 - 4 = 2$. The accidental-concurrence subset has dimension at most two, while the guaranteed component $\mathcal{S}_g$ has dimension three and dominates the union. $\square$

Remark 7.2. Proposition 7.1 is conditional: its conclusion applies only after the stated transversality hypothesis for the chosen parameter family has been verified. Under that hypothesis it is an existence mechanism, but a particular aesthetically chosen instance, such as $h = a/s$, still needs its own global certificate. This is the precise place where a Gröbner-basis or Hilbert-polynomial computation enters.

8 Canonical constraint centers and catalogue status

For $C = 1, \lambda = 1$, equivalently $h = a/s$ in (15), all six proposed centers have barycentrics

$$X_{g_i} = \left(1 + \frac{a}{s}g_i : 1 + \frac{b}{s}g_i : 1 + \frac{c}{s}g_i \right). \quad (16)$$

Because g_i is fully symmetric, these points lie on the incenter–centroid line $X(1)X(2)$: their barycentric vectors are affine combinations of (a, b, c) and $(1, 1, 1)$, with a triangle-dependent coefficient.

The normalization is part of the definition. Replacing g_i by a nonzero scalar multiple leaves its zero set unchanged but usually produces a different triangle center in (16). Thus one face constraint gives a continuum of centers, not one canonical ETC entry.

No ETC identification is asserted here. A finite failed search would not prove nonoccurrence, because catalogues evolve and projectively equivalent formulas may look very different. The six explicit functions (16) are instead proposed as natural candidates for future exact comparison, with g_1 and g_6 carrying the strongest tetrahedral significance. Any identification should be certified by clearing denominators and proving cyclic projective proportionality symbolically; catalogue recognition is not used in any theorem of this paper.

9 Algorithms and reproducibility

The calculations in this paper use two complementary methods.

Finite branch enumeration. When g is a product of explicitly listed factors, each face chooses one factor. A four-face system is obtained by taking the Cartesian fourth power of that branch list. Linear branches are settled by exact rational row reduction. For the Jaws–Tripods theorem, the human classification proves the maximal components and a finite check retains only the six irredundant alternating-path strata. The g_6 theorem needs no branch enumeration: Theorem 3.2 and the explicit geared jaws already give matching upper and lower bounds.

Gröbner dimension. For nonlinear branches, set one edge equal to 1, form the ideal over $\mathbb{Q}$, compute a Gröbner basis, and read the Krull dimension from the leading-term ideal or Hilbert polynomial. Since $\mathcal{E}$ lies in the positive chart $x_{34} > 0$, restoring scale adds one and proves the Euclidean upper bound in Proposition 5.1.

Z-cevian dimension. Given an algebraic center function F , form the six residuals (3), clear denominators, saturate by the denominators and by the Cayley–Menger boundary when appropriate, and compute the dimension of the resulting ideal. A primary decomposition distinguishes the guaranteed component $\mathcal{S}_g$ from remote accidental components. A Jacobian rank at one point certifies only a local lower-dimensional branch; it does not exclude a separate larger component.

Minimal supplement. No Mathematica notebook is required. The journal supplement consists of one short Sage script, `FaceConstraintChecks.sage`. It performs the finite rational row reductions for the all-isosceles branches, verifies the six alternating-path row spaces explicitly, and carries out the 81 exact Gröbner-dimension checks for each of $g_2, \dots, g_5$. The advertised counts are enforced by exact assertions. The dimension bounds, the g_6 theorem, the dimension-five obstruction, and the odd-dimensional local certificate are all proved directly in the text.

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10 Conclusion and further questions

Face constraints provide a direct way to manufacture spatial concurrence from planar geometry. The construction is robust: whenever a triangle center becomes the centroid on $g = 0$, the

four-face lift $\mathcal{S}_g$ is automatically contained in its Z -cevia family. The geometry is nevertheless subtler than this inclusion suggests.

The main conclusions are:

- (i) A nontrivial symmetric homogeneous real-analytic constraint imposed on all four faces produces at most a three-dimensional tetrahedral family. Thus no such $\mathcal{S}_g$ can have dimension four or five.
- (ii) A proper real-analytic Z -cevia family with nonvanishing facial weights cannot have a five-dimensional analytic component. The omitted-opposite-edge structure of the compatibility residuals is the decisive reason.
- (iii) The explicit analytic center (4) has a genuine three-dimensional jaw component, certified by an exact nonzero Jacobian minor. Thus odd-dimensional components do occur.
- (iv) For the isosceles constraint, the maximal-dimensional locus is exactly jaws and tripods. Six two-dimensional alternating-path components complete the exact classification.
- (v) The constraints $g_2, \dots, g_5$ behave generically and give dimension two.
- (vi) The 2:1 constraint g_6 gives another exact dimension-three example and contains the geometrically simple geared jaws.
- (vii) The passage from $\mathcal{S}_g$ to a particular Z -cevia family has an accidental-concurrence loophole. Under the explicit transversality hypothesis of Proposition 7.1, generic multipliers have only the expected two-dimensional accidental locus; any distinguished explicit center still requires its own global algebraic certificate.

Several problems remain natural.

- (1) Determine an aesthetically simple explicit analytic center whose entire proper Z -cevia family has global dimension exactly three, or certify that (4) already has this property.
- (2) Compute a saturated primary decomposition for the canonical centers $X_{g_1,1}$ and $X_{g_6,1}$. Does either have a remote four-dimensional component?
- (3) Determine for which ratios ρ the family $\mathcal{S}_{g_\rho}$ changes combinatorial type.
- (4) Search the full ETC symbolically for projective equivalents of (16), and, if none occur, submit the simplest dimension-three examples with their tetrahedral interpretation.
- (5) Extend Theorem 3.2 to n -simplices and constraints on m -faces. The support lemma should be replaced by a rank theorem for the incidence structure of complete uniform hypergraphs.

A Exact finite certificate

The accompanying file `FaceConstraintChecks.sage` is the complete computer-algebra supplement. It uses exact rational arithmetic. For g_1 it enumerates the $3^4 = 81$ choices of one equality per face, row-reduces their coefficient matrices, removes duplicate row spaces, and tests containment. Its output is

rank	3	4
assignments	15	66
distinct row spaces	15	24,

with 18 of the rank-four spaces contained in a rank-three space and 6 irredundant rank-four spaces. The script additionally generates the six vertex-relabelings of the alternating-path row space, asserts exact set equality with those six irredundant spaces, and prints `g1 alternating-path`

spaces verified 6. Thus the finite certificate checks the geometric identification used in the body of the paper.

For $g_2, \dots, g_5$, the same file sets $x_{34} = 1$, forms all 81 branch ideals for each constraint over $\mathbb{Q}$, and computes their Krull dimensions. The exact output is

```
g2 {1: 81}
g3 {1: 81}
g4 {1: 81}
g5 {1: 81}
```

Restoring the scale coordinate gives positive-cone dimension two for every branch meeting $\mathcal{E}$. No statement is made about components confined to $x_{34} = 0$. The script contains the factor definitions themselves; no omitted helper, external ETC table, random sampling, floating-point rank, or Mathematica notebook is needed.

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